

# Customer Churn Prediction

*A Data-Driven Retention Strategy for Telecom*

GCI World 2026 April · Final Business Proposal · Esraa

**June 2026**

Model: Tuned LightGBM | AUC: 0.698 | Recall @0.35: 90.1%

Net Annual Impact: +\$5.5M | ROI: 5.6× | Dataset: 100,000 customers

# Agenda

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## 01 **Market Context & Business Problem**

Why churn costs telecom firms billions

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## 02 **Exploratory Data Analysis**

Key patterns discovered in 100 k customer records

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## 03 **Feature Engineering**

12 engineered signals from raw data

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## 04 **Model Selection & Results**

Tuned LightGBM · AUC 0.698 · Recall 90.1 %

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## 05 **Risk Segmentation**

Three actionable tiers validated against actual churn

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## 06 **Business Proposal & ROI**

Retention program worth +\$5.5 M net annual impact

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## 07 **Conclusion & References**

Next steps and citations

# The Churn Problem Is a Billion-Dollar Issue

Market context

**\$29B+**

Annual industry revenue lost to customer churn (GSMA 2024)<sup>1</sup>

**5–7×**

Cost to acquire vs. retain a single customer<sup>2</sup>

**2.2–2.8%**

Monthly churn rate — US telecommunications sector<sup>3</sup>

## Why Prediction Changes the Economics

- Reactive retention (calling already-churned customers) is too late — and costly.
- Blanket discount campaigns waste budget on customers who would have stayed anyway.
- A predictive model lets the firm intervene only on the right customers at the right time.
- Our model identifies 90.1 % of churners before they leave — enabling targeted outreach.

# EDA — Dataset & Class Balance

100,000 customer records · nearly balanced classes

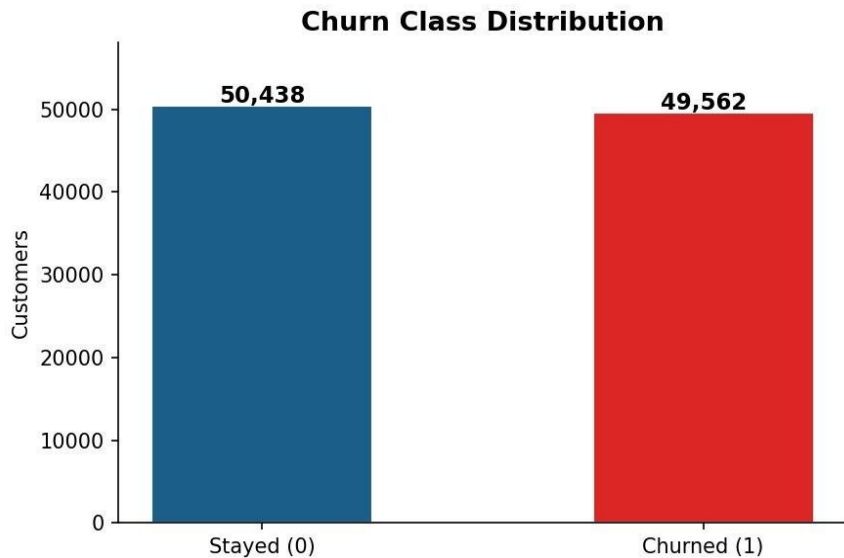


Figure 1 — Churn class distribution (balanced dataset)

## Dataset at a Glance

- 100,000 customers | ~230 raw features
- Stayed: 50,438 (50.4 %) Churned: 49,562 (49.6 %)

## Key EDA Findings

- Classes nearly balanced — no resampling required
- 10 high-missingness columns (>20 %) identified as drop candidates
- Churners show higher equipment age ( $\mu=421$  d vs 363 d for stayers)
- Refurbished-device customers churn at ~53 % vs ~49 % for new-device

# EDA — Data Quality & Behavioral Signals

Missing-value audit + usage & service metrics

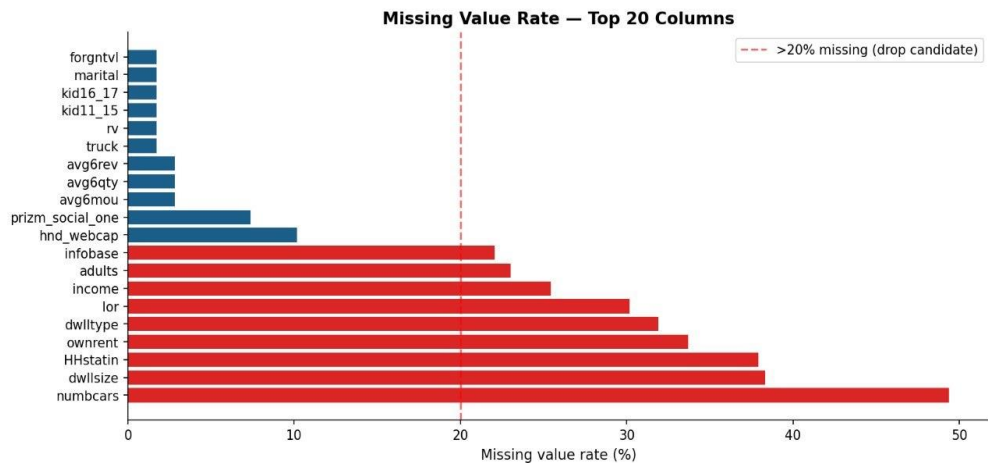


Figure 2 — Top 20 columns by missing-value rate; red = >20% threshold (dropped)

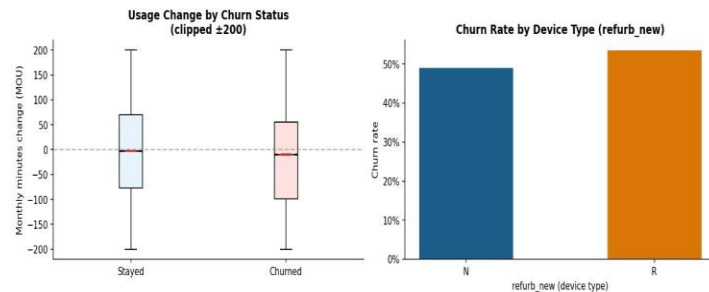


Figure 3 — Usage change & device-type churn rate

## Takeaways

- Demographic cols dropped (numbcars, dwllsize, etc.)
- Churners show steeper MOU decline (wider IQR)
- Refurbished device → higher churn (device dissatisfaction signal)
- Service-quality signals (drop calls, blocked calls) differ minimally

# EDA — Equipment Age: The #1 Predictive Signal

Churners carry older devices on average

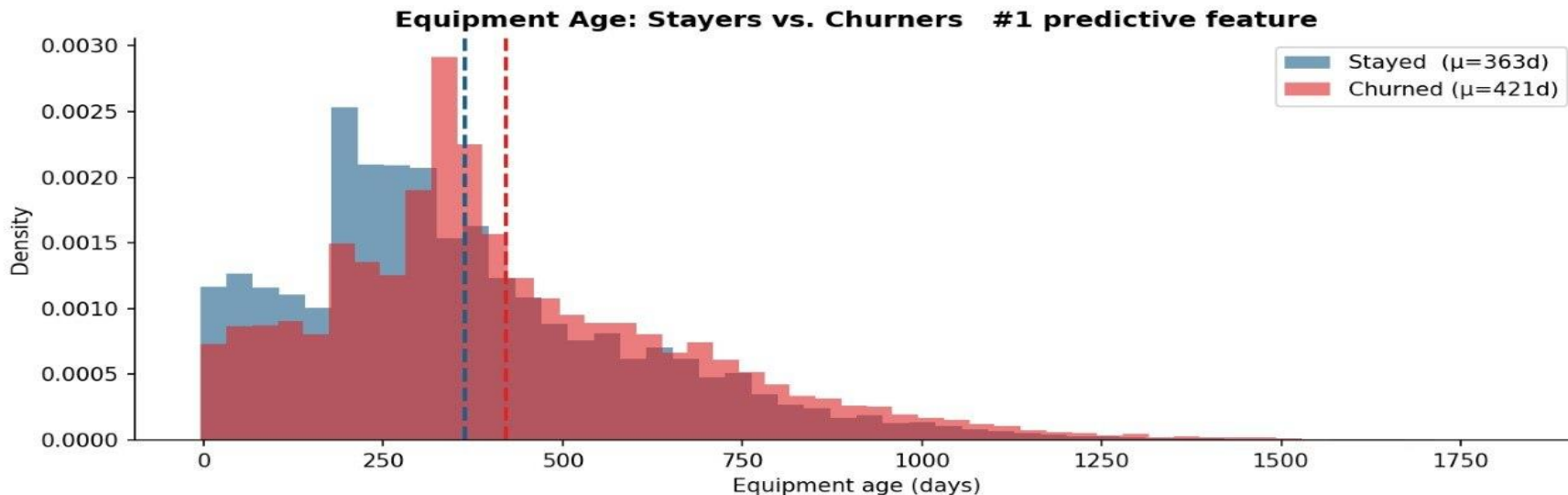


Figure 4 — Equipment age distribution: stayers ( $\mu=363$  d) vs churners ( $\mu=421$  d)

**+58 days**

Average equipment-age gap  
between churners and stayers

## Interpretation

Older equipment correlates with upgrade  
desire and contract expiry — prime churn  
moments.

## Business Implication

Proactive upgrade offers timed to device age  
can neutralize the single strongest churn  
driver.

# Feature Engineering — 12 Engineered Signals

Transforming raw telecom data into predictive features

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## rev\_per\_min

Revenue efficiency —  $\text{Total revenue} \div \text{minutes-of-use}$

## mou\_rev\_sync

Usage-revenue alignment — Whether revenue and usage move together

## change\_rev

Revenue momentum — Month-over-month revenue delta

## abs\_change\_mou

Usage volatility — Absolute MOU change — erratic usage flags risk

## usage\_momentum

Trend direction — Rolling MOU trend — declining = warning signal

## call\_completion\_rate

Network quality proxy —  $\text{Completed} \div \text{attempted calls}$

## failure\_rate

Call failure rate —  $\text{Failed} \div \text{total calls}$  (service dissatisfaction)

## totmrc\_Mean

Contract value — Mean monthly recurring charge (agg. across months)

## mou\_Mean / mou\_cvce\_Mean

Usage profiles — Average & peak-hour usage aggregates

## inonemin\_Mean

Roaming exposure — International-one-minute call behavior

## total\_value\_per\_month

ARPU proxy — Revenue per active month

## eqpdays

Device age — Equipment age in days — top-ranked feature

# Model Selection — Why LightGBM?

Comparing approaches for imbalanced tabular classification

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## Logistic Regression

Baseline

Simple, interpretable, but linear — misses non-linear churn signals.

## Random Forest

Ensemble

Strong on tabular data but slower; less efficient on 230+ features.

## LightGBM v3 (hand-picked)

AUC 0.6939

Gradient boosting with leaf-wise growth; efficient and powerful.

## LightGBM v3 (ensemble ×4)

AUC 0.6980

Best of 4 seeds — improved but complex to deploy.

## LightGBM v4 + Optuna (30 trials)

AUC 0.6967 ✓ FINAL

Optuna hyperparameter tuning: reproducible, single model, excellent recall.

# Model Results — Feature Importance & Tuning

Top signals and Optuna hyperparameter search

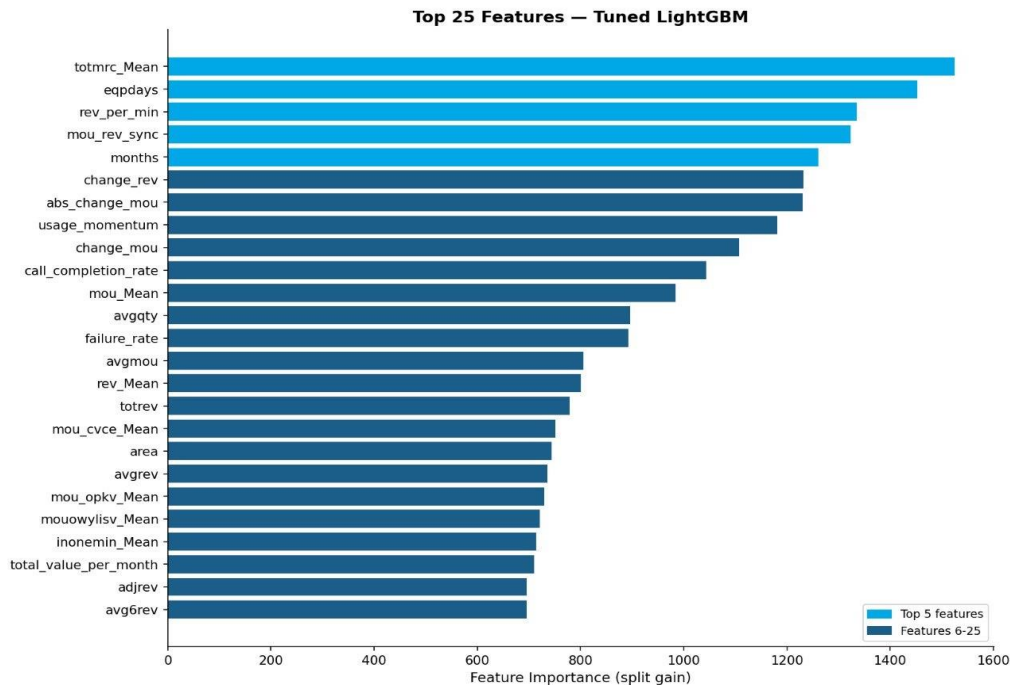


Figure 5 — Top 25 features (split gain), Tuned LightGBM

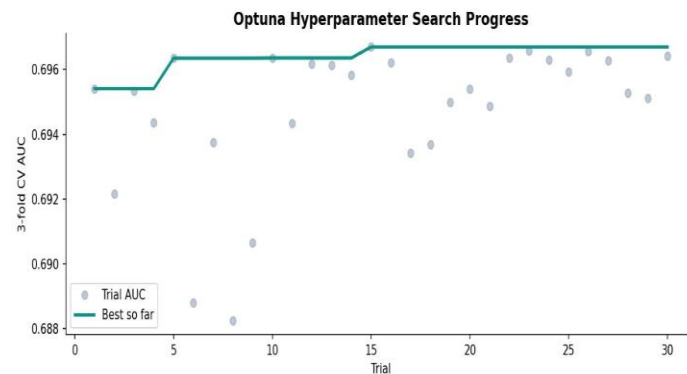


Figure 6 — Optuna 30-trial search; best OOF AUC 0.6967

## Top 5 Predictive Features

- totmrc\_Mean — Monthly recurring charge avg.
- eqpdays — Equipment age (days)
- rev\_per\_min — Revenue efficiency
- mou\_rev\_sync — Usage-revenue alignment
- months — Customer tenure

# Model Evaluation — Performance Metrics

Tuned LightGBM · Test set · Decision threshold 0.35

Model: Tuned LightGBM (Optuna, 30 trials) | Metric: AUC-ROC | Score: 0.698 (test) | OOF AUC: 0.6967

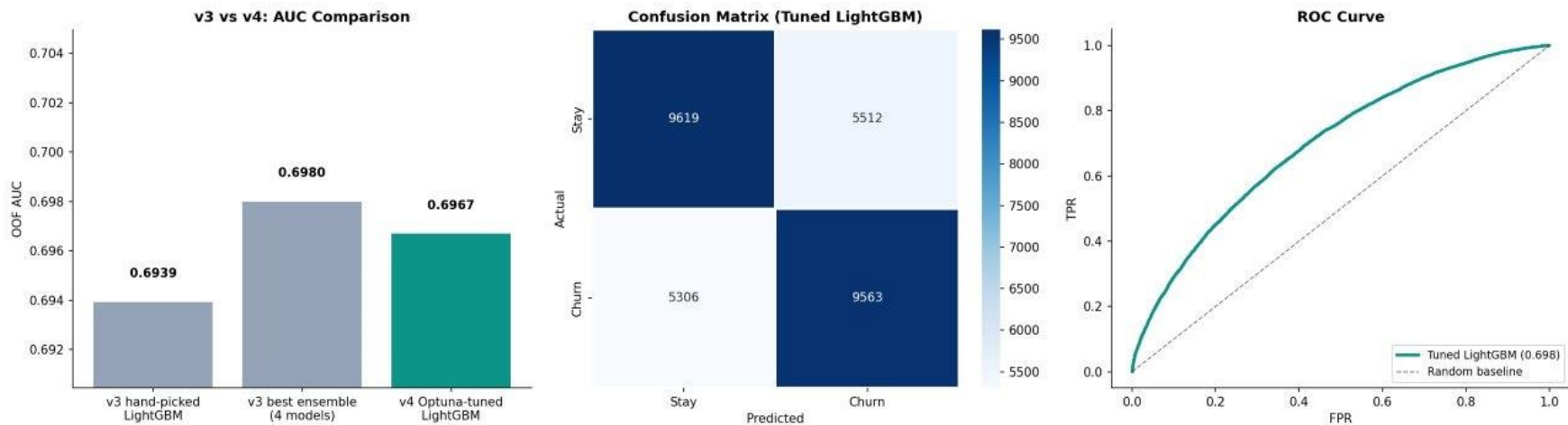


Figure 7 — AUC comparison (v3 vs v4), Confusion Matrix, ROC Curve

0.698

Test AUC-ROC

90.1%

Recall @ thresh 0.35

56%

Precision @ thresh 0.35

9,619

True Negatives (stay correct)

# Threshold Selection — Recall vs. Precision Trade-off

Lower threshold catches more churners at the cost of more false positives

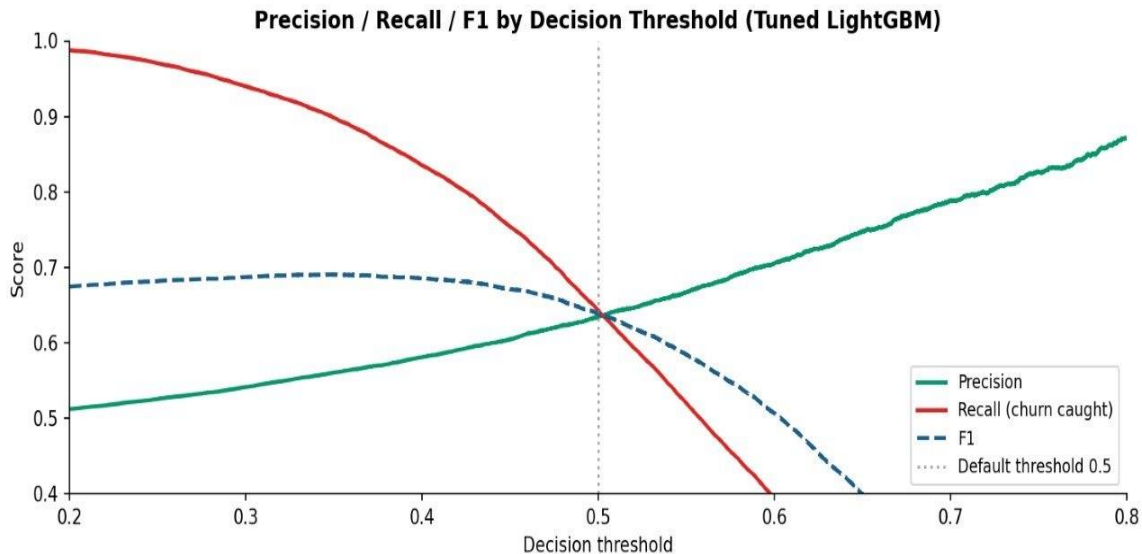


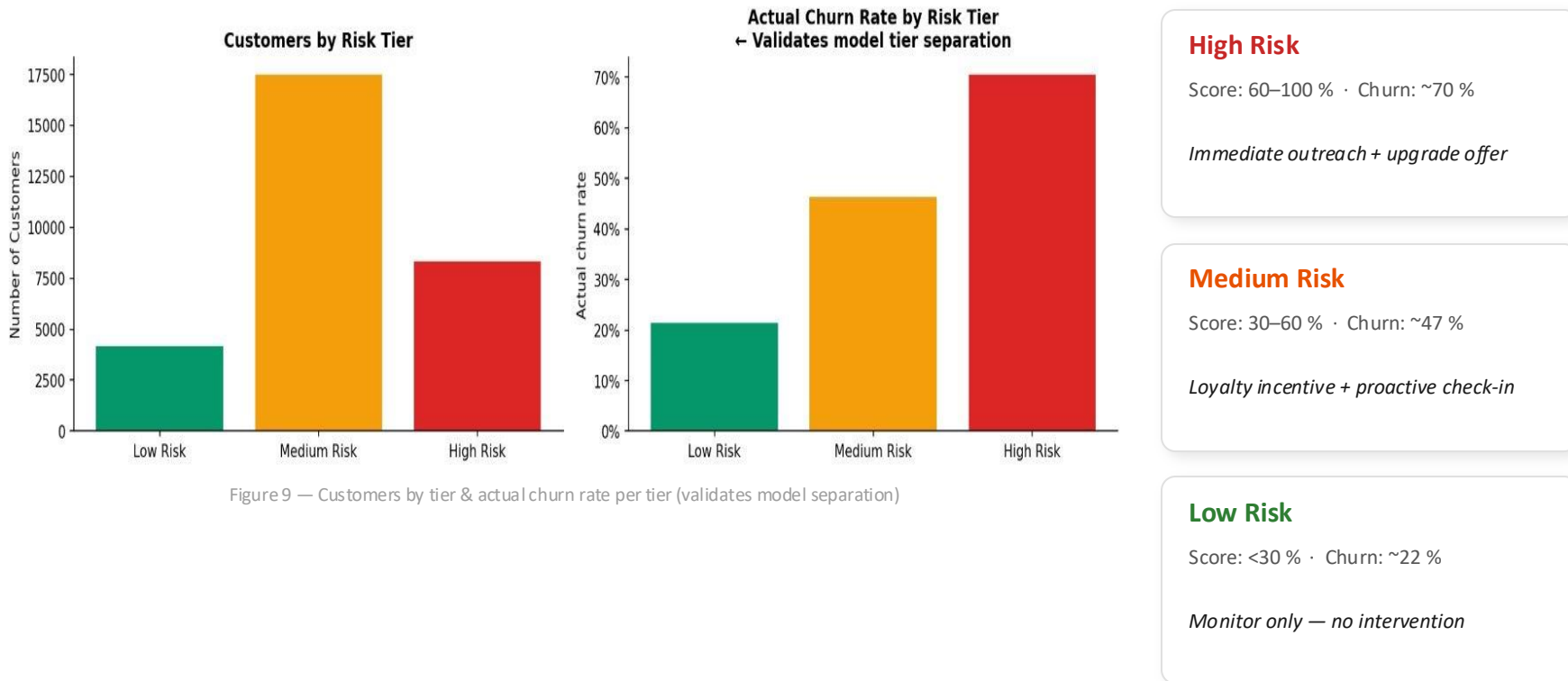
Figure 8 — Precision/Recall/F1 by decision threshold (Tuned LightGBM)

## Why 0.35 Threshold?

- Recall = 90.1 % at threshold 0.35 — we catch 9 in 10 churners before they leave.
- False-positive cost (unnecessary outreach) is far lower than false-negative cost (lost customer).
- F1 peaks near 0.5 but the business objective is minimizing missed churners.
- Sensitivity analysis confirms positive net impact down to 8% offer success rate.

# Risk Segmentation — Three Actionable Customer Tiers

Model scores translated into operational intervention segments



# Business Proposal — Targeted Retention Program

Data-driven intervention with quantified financial impact

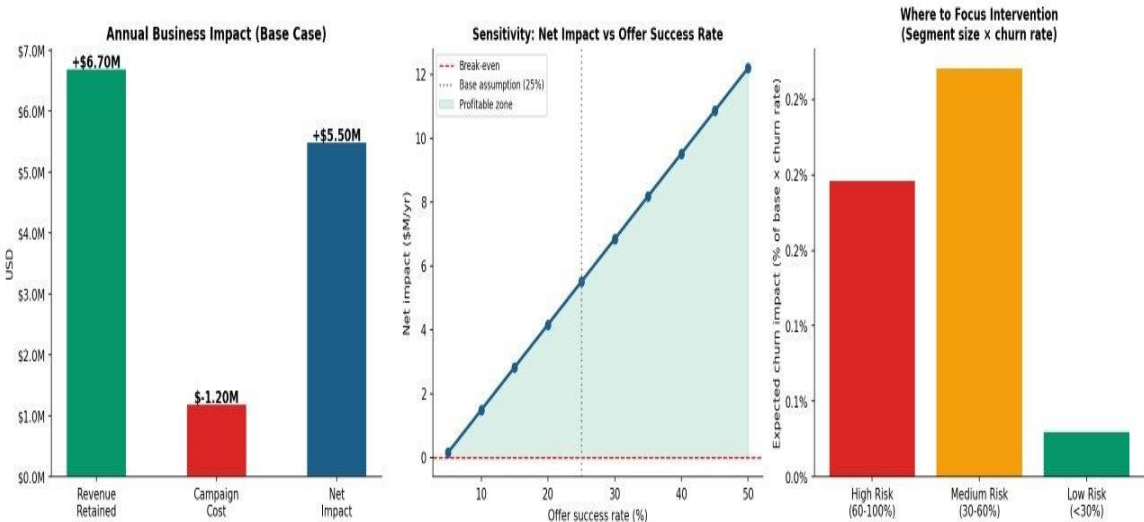


Figure 10 — Annual impact (base case), sensitivity analysis, and intervention priority

## ROI Summary

|                   |                    |
|-------------------|--------------------|
| Revenue Retained  | +\$6.70M           |
| Campaign Cost     | -\$1.20M           |
| Net Annual Impact | +\$5.50M           |
| ROI               | 5.6×               |
| Break-even rate   | ~8 %               |
| Assumption        | 25 % offer success |

Assumptions: avg. monthly revenue \$28/customer · retention offer cost \$48/customer · 25% success rate

# Key Takeaways

**90.1%**

of churners identified  
before they leave

**+\$5.5M**

net annual impact  
at 25% offer success

**5.6×**

return on investment  
from retention spend

**3 tiers**

for targeted, efficient  
intervention

## Next Steps

- Deploy model to production CRM and score all customers weekly
- A/B test retention offer (upgrade voucher vs. loyalty discount) on High-Risk segment
- Monitor actual vs. predicted churn monthly; retrain every quarter
- Explore SHAP explanations for per-customer interpretability in agent dashboards

# References

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